

Data Engineering on Google Cloud — Student Pack

Welcome to your two-day course. This pack is everything handed to you directly: schedules, module notes, lab companions, and the reference cards you'll want back at your desk.

The course at a glance

Course	Data Engineering on Google Cloud
Dates	Monday October 5 – Tuesday October 6, 2026
Time	9:00 AM–4:30 PM CT (lunch 12:00–1:00, morning and afternoon breaks)
Format	Virtual, instructor-led, hands-on throughout
Join	Google Meet — both days
Labs	Google Skills classroom

Joining, check-in, and ground rules

- **Join the Meet** at 9:00 AM CT both days: <https://meet.google.com/ogk-ivdv-bhi>. Tip: if you're signed into several Google accounts, run the Meet in an incognito window and the labs in your normal one (or vice versa) — multi-account sign-in can kick you out.
- **Check in for Google** when you arrive: [Google check-in form \(QR code\)](#) — Google keeps an attendance record for the class.
- **No recording.** Recording is not permitted for this course (course copyrights) — the pack you're holding is the take-away.
- **Lab window.** Google Skills labs open **Monday Oct 5 at 6:00 AM CT** and close **Tuesday at 11:00 PM CT**. Everything in a lab project disappears when the lab ends — if something matters, screenshot it.
- **Free credits after class.** You can request [50 Google Skills On-Demand credits per training day](#) — they arrive within a week and last 30 days. Perfect for re-running the Datastream and Dataform labs back at your desk.
- **Course evaluation.** At the end of Day 2, please take 3 minutes for the [event evaluation](#) — it's how courses like this get tuned for the next team.

Before class: tell us who's in the room (5 minutes)

Fill in **your row** in the [pre-course survey](#) (anyone with the link can edit — no sign-in needed). Your name is already there. Rate yourself 0–3 on the tools you use today, and — most valuable of all — name **one real extract job you own**. That job becomes your group's raw material for the Day-2 capstone, where you map it onto the GCP target architecture. This calibrates the pace of the two days; it is not a quiz.

Who this course is for

You — the Iowa DOM DoIT data team. Roughly ten SQL Server/SSIS ETL developers, the M&O team that keeps the jobs running, and the Power BI report authors the state relies on. You're senior in data engineering: ETL and ELT, change data capture, scheduling, source control, CI/CD pipelines for SQL Server deployments. What you're new to is the Google Cloud console — so these two days map what you already know onto where things live in GCP, instead of re-teaching concepts you own.

Your instructor: Imran Ahmad, PhD — Google Cloud Authorized Trainer and certified Professional Data Engineer, who builds large-scale production data pipelines for the Canadian federal government and writes the books on AI agents. See the [instructor profile](#).

The course is built around your real project:

- Retire the legacy nightly extracts.
- Migrate the SQL Server data warehouse to BigQuery as the single source of truth.
- Build a Bronze → Silver → Gold medallion lakehouse.
- Use Dataform for the ELT work between Silver and Gold.
- Feed your existing Power BI reports from the Gold data marts.

How the two days fit your migration

Day 1 — Landing your data in BigQuery. Console confidence first, then data in: four ways to load files, a live replication stream replacing the nightly extract window, and Dataform — your Silver→Gold tool. You end the day by scheduling your first query. See the [Day 1 schedule](#).

Day 2 — Your target operating model: lakehouse & medallion. Lakes vs. warehouses vs. lakehouses, external tables and Apache Iceberg (Bronze made tangible), governance and Google's own Bronze/Silver/Gold architecture, the Power BI last mile, and a capstone where you map one of your real extract jobs onto the GCP design — an artifact you take back to work. See the [Day 2 schedule](#).

How the labs work

- Labs run in the [Google Skills classroom](#). Open the classroom, sign in, and launch the lab when it's called.
- Each lab gives you a **temporary GCP project** with real credentials. When you start a lab, **the lab timer starts** — watch it. When time expires, the project is deleted. Nothing in a lab project is permanent, and that's by design: it's a disposable sandbox.
- Lab projects are safe to break. You cannot damage anything that matters.
- The lab's own step-by-step instructions live in the classroom. The companions in this pack add orientation, gotchas, and context around them — they don't replace the lab instructions.

- One thing you create yourself (not in a lab): on Day 1 morning you create a dataset named class in your lab project. Every afternoon loading activity writes into it — don't skip that step.

What's in this pack

Folder	What you'll find
day1/	Day-1 schedule, a notes file per teaching block, and companions for every hands-on activity and lab
day2/	Same treatment for Day 2
reference/	Keep-forever reference cards (listed below)

Every module folder also contains the **slide deck** for that module (slides.pdf), and every document comes in both **Word (.docx)** and **PDF** form — use whichever you prefer; the content is identical.

The reference cards:

- [GCP for SQL Server professionals](#) — the big mental-model mapping card
- [T-SQL → BigQuery SQL](#) — dialect cheat sheet
- [BigQuery console survival guide](#) — where everything lives, for first-time console users
- [Migration map](#) — your end-to-end target architecture on one page
- [Power BI ↔ BigQuery connector guide](#) — the last mile to your existing reports
- [Glossary](#) — the GCP vocabulary in SQL Server terms

Ground rules

- **Ask anything.** You're senior engineers who are new to one console. "Where do I click?" questions are expected and welcome — sixteen people learning one new UI goes faster when nobody sits stuck.
- **Parking lot.** Questions that would pull the room off the day's path get written down visibly and answered before the day ends.
- **Labs are safe to break.** Temporary projects, free or pennies of cost, nothing connected to state systems. If something fails, that's a learning moment — say so and we'll look at it together.
- **This pack is yours.** Take it with you. The reference cards were written to be useful at your desk the week after class, not just during it.